
2.4 Course Content:

1. **Introduction to Artificial intelligence:** What is AI, History of AI, Types of AI, Components of Intelligence, Agent and Environment, Informed Search Algorithms: Pure Heuristic, Best first, A*, AO* Algorithms.
2. **Knowledge Representation:** Introduction, Knowledge representation, Knowledge based agent, types of knowledge, Techniques of knowledge representation in AI, Syntax.
Reasoning and Learning in AI: Reasoning, Probabilistic Reasoning in AI.
3. **Introduction to Machine Learning**
 - **Introduction:** Need for Machine Learning, Machine Learning Explained, Machine Learning in Relation to other Fields, Types of Machine Learning.
 - **Understanding Data:** Introduction, Univariate Data Analysis and Visualization. Bivariate Data and Multivariate Data, Feature Engineering and Dimensionality Reduction Techniques.
4. **Basic Learning Theory:** Design of Learning System, Introduction to Concept of Learning, Modelling in Machine Learning.
Similarity-based Learning: Nearest-Neighbor Learning, Weighted K-Nearest-Neighbor Algorithm.
Regression Analysis: Introduction to Regression, Introduction to Linear Regression, Multiple Linear Regression.
Decision Tree Learning: Introduction to Decision Tree Learning Model, Decision Tree Induction Algorithms: ID3 Tree Construction.
5. **Bayesian Learning:** Introduction to Probability-based Learning, Fundamentals of Bayes Theorem and Classification Using Bayes Model: Naïve Bayes Algorithm.
Clustering Algorithms: Introduction to Clustering Approaches, K-means Algorithm
Artificial Neural Networks: Introduction, Biological Neurons, Artificial Neurons, Perceptron and Learning Theory: XOR, Types of Artificial Neural Networks.

Laboratory Programs:

1. Design and implement the A* search algorithm for a given weighted graph. The algorithm should compute the shortest path from a start node to a goal node using:
 $g(n)$: Cost from start node to current node
 $h(n)$: Heuristic estimate from current node to goal
 $f(n) = g(n) + h(n)$.

